STATIC INCENTIVES ⟶ COLLECTIVE KNOWLEDGE

Better advice, weaker learning?

AI, Human Cognition and Knowledge Collapse

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github.com/gsaco/ai-04-acemoglu

Five-minute discussion. Focus: static problem and Observation 1.

1 The paper and the agent's problem

Can better individual advice weaken the knowledge everyone needs?

$$\max_{e\geq 0} \; U(e; X, \tau_A) = G(X)\Delta_G + G(X)G(Y)\Delta_X - \frac{e^\alpha}{\alpha}, \qquad Y = \sigma^{-2} + \lambda_I e + \tau_A.$$

- X: inherited public precision; Y: context-specific precision.
- $G(z) = 2\Phi(\sqrt{z}) - 1$: probability the prediction error is within 1.
- Effort produces private information and a public spillover; short-lived, atomistic agents take X as given.

Production restriction: $\Delta_I = 0$, $\Delta_X > 0$, $\Delta_G + \Delta_X = 1$.

Source: §§3.1–3.3, pp. 7–12, eqs. (4)–(6); constant $f(0,0)$ omitted.

The supplied paper's §2 is related literature; its static model starts in §3.

2 Observation 1: two signs, explicit conditions

$$g(z) = \frac{\phi(\sqrt{z})}{\sqrt{z}}, \qquad g'(z) = -\frac{1}{2}(1 + z^{-1})g(z) < 0.$$

$$\underbrace{U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0}_{\text{general knowledge complements effort}} \qquad \underbrace{U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0}_{\text{agentic advice substitutes for effort}}$$

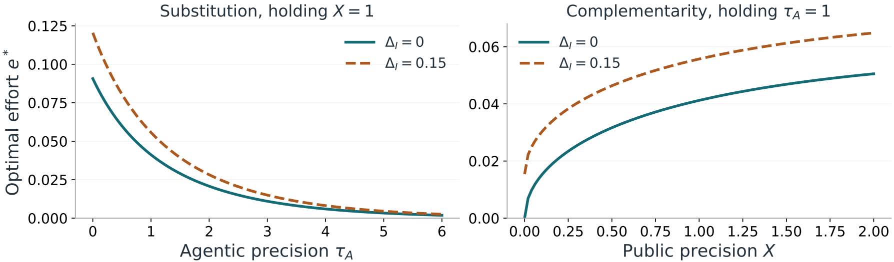
Conditions: $X > 0$, finite $\tau_A \geq 0$, $\sigma^{-2} > 0$, $\lambda_I > 0$, $\alpha > 1$; Assumption 1, independent Gaussian information, unit-tolerance payoffs; public learning is not privately internalized.

$U_{ee} < 0$ gives a unique interior optimum: $e^*_X > 0$, $e^*_{\tau_A} < 0$. At $X = 0$, $e^* = 0$; the strict interior statements do not apply.

Welfare caution: fixed-X utility rises; long-run welfare need not. The single-peak claims require $\sigma^{-2} \geq \sqrt{2} - 1$ and the relevant $\alpha - 1 \gtrless 1/4$ regime; the optimum may be $\tau^*_A = 0$.

Source: Observation 1, p. 13; Observation 2, p. 14; Propositions 10–11, p. 26.

3 What I did: verify signs, test the hidden premise

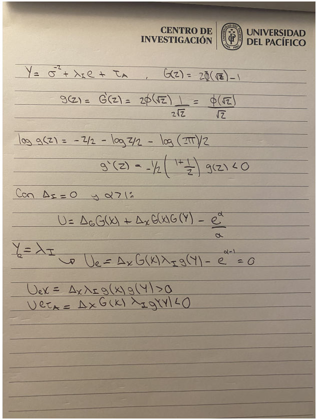


Reproducible static checks: SymPy identities + 444 numerical optima; maximum FOC residual $< 2.5 \times 10^{-15}$.

Extension: allow $\Delta_I = 0.15$. Both signs survive, but at $X = 0$, $\tau_A = 1$, effort rises from 0 to 0.01539.

Own AI-assisted analysis in extra/static_checks.py; $\alpha = 2$, $\lambda_I = 1$, $\sigma^{-2} = 1$, $\Delta_X = .6$; $\Delta_G = 1 - \Delta_I - \Delta_X$. No dynamic results simulated.

4 Where I challenged the AI's reasoning



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$Y = \sigma^{-2} + \lambda_I e + \tau_A$, $G(z) = 2\Phi(\sqrt{z}) - 1$

$g(z) = G'(z) = 2\phi(\sqrt{z}) \frac{1}{2\sqrt{z}} = \frac{\phi(\sqrt{z})}{\sqrt{z}}$

$\log g(z) = -z/2 - \log z/2 - \log(2\pi)/2$

$g'(z) = -\frac{1}{2}\left(1 + \frac{1}{z}\right) g(z) < 0$

Con $\Delta_I = 0$ y $\alpha > 1$:

$U = \Delta_G G(X) + \Delta_X G(X) G(Y) - \frac{e^\alpha}{\alpha}$

$Y_e = \lambda_I \rightarrow U_e = \Delta_X G(X) \lambda_I g(Y) - e^{\alpha-1} = 0$

$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0$

$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0$

Claim tested: "More accurate AI must raise welfare."

Verdict: true for optimized static utility at fixed X; not a general long-run conclusion.
Public-knowledge losses also matter.

The assumption audit: §5 relaxes aggregation, synthetic data and effort bundling. It retains $\Delta_I = 0$.

$$U_e\big|_{X=0,e=0} = \Delta_I \lambda_I g(\sigma^{-2} + \tau_A) > 0$$

My extension: with $\Delta_I > 0$, private learning remains valuable at $X = 0$.

Source: Assumption 1, p. 8; §4.3, pp. 24–26; §5, pp. 29–32.